

Table 3.8.1.3-1—Wind Load Components on Live Load

Skew Angle (degrees)	Transverse Component (klf)	Longitudinal Component (klf)
0	0.100	0.000
15	0.088	0.012
30	0.082	0.024
45	0.066	0.032
60	0.034	0.038

For the usual girder and slab bridges having an individual span length of not more than 150 ft and a maximum height of 33.0 ft above low ground or water level, the following wind load components on live load may be used:

- 0.10 klf, transverse
- 0.04 klf, longitudinal

Both forces shall be applied simultaneously.

3.8.2—Vertical Wind Load

The effect of forces tending to overturn structures, unless otherwise determined in Article 3.8.3, shall be calculated as a vertical upward wind load equal to:

- 0.020 ksf for Strength III load combination and
- 0.010 ksf for Service IV load combination

times the width of the deck, including parapets and sidewalks, shall be applied as a line load. This force shall be applied only when the direction of horizontal wind is taken to be perpendicular to the longitudinal axis of the bridge. This line load shall be applied at the windward quarter-point of the deck width in conjunction with the horizontal wind loads specified in Article 3.8.1.

The vertical wind load shall not be applied for load combinations other than Strength III and Service IV.

3.8.3—Wind-Induced Bridge Motions

3.8.3.1—General

The provisions of this Article shall apply to bridges in service and to bridges during construction after the deck and all features affecting their aeroelastic behavior are installed.

Force effects of wind-induced vibrations shall be taken into account in the design of bridges and structural components apt to be wind-sensitive. For the purpose of this Article, the following bridges shall be deemed to be wind-sensitive:

- all bridges with a span-to-depth ratio, and structural components thereof with a length-to-width ratio, exceeding 30.0,

C3.8.2

The intent of this Article is to account for the effect resulting from interruption of the horizontal flow of air by the superstructure. This load is to be applied even to discontinuous bridge decks, such as grid decks. This load may govern where overturning of the bridge is investigated.

For flexible bridges, such as cable-stayed and suspension bridges, the vertical force should be checked as an upward or downward force, whichever may control the design.

C3.8.3.1

Because of the complexity of analyses often necessary for an in-depth evaluation of structural wind-induced vibrations, this Article is intentionally kept to a simple statement. Many bridges, decks, or individual structural components have been shown to be insensitive to wind-induced vibrations if the specified ratios are under 30.0, a somewhat arbitrary value helpful only in identifying likely wind-sensitive cases.